Homework 4

PHYS 6020 - Fall 2023

Due on Friday, September 22nd 2022

Problem 1

Let $\hat{H}$ be an Hermitian operator (that may represent the Hamiltonian of an arbitrary physical system) and $\{|\phi_j\rangle\}$ its basis vectors with eigenvalues E_j. The operator $\hat{U}(m,n)$ is defined by:

$$\hat{U}(m,n) = |\phi_m\rangle\langle\phi_n| \tag{1}$$

(i) Calculate the adjoint $\hat{U^{\dagger}}(m,n)$

(ii) Calculate the commutator $[\hat{H}, \hat{U}(m,n)]$

(iii) Prove the relation

$$\hat{U}(m,n)\,\hat{U^{\dagger}}(p,q) = \delta_{nq}\,\hat{U}(m,p) \tag{2}$$

(iv) Calculate $\mathrm{Tr}\,\hat{U}(m,n)$, where Tr stands for trace

(v) We define the operator $\hat{A}$ with matrix elements $A_{nm} = \langle\phi_n|\hat{A}|\phi_m\rangle$. Prove the relation

$$\hat{A} = \sum_{m,n} A_{mn}\,\hat{U}(m,n) \tag{3}$$

(vi) Show that $A_{pq} = Tr[\hat{A}\,\hat{U^{\dagger}}(p,q)]$

Problem 2

Consider a two-dimensional vector space spanned by the basis vectors $\{|1\rangle, |2\rangle\}$ and the operator:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{4}$$

(i) Is σ_y Hermitian? Calculate its eigenvalues and eigenvectors with components given in terms of the basis $\{|1\rangle, |2\rangle\}$.

(ii) Calculate the matrices that represent the projectors onto these eigenvectors. Do they satisfy the orthogonality and completeness relations?

(iii) Same questions for the matrices:

$$A = \begin{pmatrix} 2 & i\sqrt{2} \\ -i\sqrt{2} & 3 \end{pmatrix} \tag{5}$$

for the two-dimensional space, and

$$L_y = \frac{\hbar}{2i} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 0 \end{pmatrix} \tag{6}$$

for the three-dimensional extension.

Problem 3

The σ_x matrix is defined by:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{7}$$

(i) Prove the relation

$$e^{i\alpha\sigma_x} = \mathcal{I}\cos\alpha + i\sigma_x \sin\alpha \tag{8}$$

where $\mathcal{I}$ is the 2×2 unit matrix.

(ii) Find an analogous relation for the matrix σ_y defined in the previous problem.

(iii) Generalize your findings for all matrices of the form

$$\sigma_u = \lambda\sigma_x + \mu\sigma_y \tag{9}$$

with $\lambda^2 + \mu^2 = 1$.